

Household Financial Transaction Data

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Abstract

The growth of the availability and use of detailed household financial transaction micro data has dramatically expanded the ability of researchers to understand both household decision making and aggregate economic fluctuations across a wide range of fields. This class of transaction data is derived from a myriad of sources, including financial institutions, FinTech apps, and payment intermediaries. We review how these detailed data have been utilized in finance and economics research and analyze both their benefits and limitations as compared to more traditional measures of income, spending, and wealth. Finally, we discuss the future potential of this flexible class of data in firm-focused research, real-time policy analysis, and macro statistics.

1. INTRODUCTION

Detailed measurement of consumption expenditures is a key ingredient for questions spanning a range of fields across economics and finance. To address such questions, researchers have typically turned to government surveys of households, used scanner data with product-level consumer good purchases, or imputed consumption as the residual of the budget constraint using administrative tax records. Each of these approaches has disadvantages, such as high costs, small sample sizes, or short panels; they may also be slow to update, cover only portions of the household budget, or be prone to measurement error.

Recent years have seen a surge in research taking advantage of a new class of data that is centered around comprehensive panels of financial transactions from credit, debit, and checking accounts across many thousands or even millions of individuals. Derived from banks, financial aggregator services (e.g., FinTech apps), or credit card companies, the data often cover high frequency financial flows and balances across many types of accounts. Researchers have employed these data to answer questions about consumption behavior, household balance sheets, and the response to income fluctuations or government policy changes, thereby greatly enhancing our understanding of heterogeneity in income, balance sheets, beliefs, preferences, and household decision making in general.

Household financial transaction data have seen a swift uptake among researchers due to the host of advantages they offer relative to existing survey, scanner, or administrative data. Massive sample sizes, often in the millions of individuals, provide increased statistical precision and the ability to drill down on small subsamples of interest. Moreover, the granularity of the data and the fact that they are often paired with other financial accounts owned by the individual mean that questions can be asked about the entirety of the household balance sheet or financial flows. Researchers can better control for a variety of confounding variables and reject alternative explanations for a given question.

As one example of the rapid proliferation of research in this area, **Figure 1** displays the trend in the number of papers utilizing transaction or payments data over time in two of the major repositories for economics and finance research: the National Bureau of Economic Research

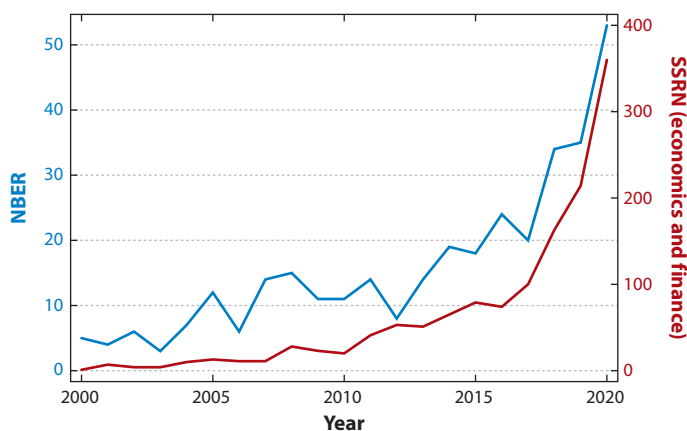


Figure 1

Counts of papers employing transaction data by year. Yearly counts were obtained from searches of papers containing the terms “credit card data,” “bank account data,” “transaction data,” and “transaction-level data” from the SSRN and the NBER working paper series from 2000 to 2020. The searches were conducted on the full text of the articles. Searches in the SSRN database were limited to economics and finance papers. Abbreviations: NBER, National Bureau of Economic Research; SSRN, Social Science Research Network.

(NBER) working paper series and the Social Science Research Network (SSRN) working paper repository. In both cases, there has been a steady growth in papers utilizing this class of data over the past two decades. Notably, 2020 saw the most dramatic growth in papers written using transaction data, as researchers took advantage of the granularity of these data and the speed with which they can be accessed to quickly write high-impact papers about the economic impacts of the COVID-19 pandemic on households and businesses. This new type of data also allowed researchers to easily evaluate some of the fiscal, monetary, and regulatory interventions enacted all over the world.

Although such granular financial data have numerous benefits, they also have limitations when compared to more traditional sources of household income and spending data such as surveys. As transaction data have increased in availability, researchers have worked to reduce their inherent disadvantages by linking transaction databases with external credit records, conducting surveys of users, or matching to administrative data. Such external linkages increase the ability of researchers to gain qualitative data about a household's characteristics, beliefs, and financial resources.

The remainder of the paper is organized as follows. Section 2 describes the sources of transaction data as well as common linkages made to transaction-level data. Section 3 compares transaction-level data to some alternative sources of consumption or household financial flow data. Section 4 details some of the research that has taken advantage of transaction-level data. Section 5 discusses emerging strands of research utilizing transaction-level data across a number of fields. Section 6 concludes.

2. SOURCES OF TRANSACTION DATA

2.1. Brokerage Accounts

A first wave of economic research using direct access to consumers' financial transactions was built around individual brokerage accounts. A range of papers focused on behavioral finance questions about investors, such as those by Odean (1998), Barber & Odean (2000, 2001), and Choi et al. (2002). With transaction-level brokerage data, researchers could better understand how retail investors made trading decisions, uncovering a number of important stylized facts about heterogeneity across observable types (e.g., gender or trading propensities), and how many investors seemed to make systematic mistakes that reduced their overall returns.

However, although these brokerage data cover one aspect of household financial transactions, researchers were unable to observe income or spending transactions among investors outside of their brokerage accounts.

2.2. Banks, Financial Intermediaries, and FinTech

Beginning in the early 2010s, detailed transaction-level data from consumers' bank accounts and credit cards began to emerge. These new data sets were supplied by a multitude of sources and have expanded to include a wide range of financial accounts beyond brokerage accounts.¹

One major source of contemporary transaction data is banking institutions. Researchers around the world have been able to form partnerships with both brick-and-mortar banks (e.g., Ganong & Noel 2019, Andersen et al. 2020b, Aydin 2022) and online banks (e.g., D'Acunto et al. 2019, Loos et al. 2020, Baker et al. 2022b) to gain access to anonymized transaction-level data from their customers. Most banks offer a wide range of services to customers, such as savings and checking

¹However, some earlier work utilized bank and credit card data aggregated to individual-month levels (see, e.g., Agarwal et al. 2007).

accounts, debit and credit cards, and mortgages and other lending facilities. Given this range of services, these partnerships offer the possibility of observing not only comprehensive spending and income flows but also various types of assets, borrowing behavior, and other demographic information collected by the banks. Researchers are sometimes limited to a subset of types of financial accounts or to transactions from a limited sample of the bank's customers.

A more limited option has been to obtain access to transaction data directly from credit card firms (e.g., Einav & Klenow 2021) or credit card issuers (e.g., Gathergood et al. 2021). Such sources are limited to spending through particular channels and cards, and likely they have no information regarding income for a given individual or household. Even narrower sets of transactions have been obtained from other proprietary sources. Cookson (2018) uses transaction-level ATM withdrawal information obtained from casinos throughout the United States. In addition, the rise of payment apps such as Apple Pay, Venmo, and AliPay has created a new avenue to directly observe household spending transactions across large portions of the population in many countries.²

While long-standing financial institutions have provided valuable access to transaction-level data, much of the pioneering work with this class of data has been enabled by an emerging ecosystem of FinTech products and apps. These products take two extremely broad forms: those that act mainly to aggregate and track a customer's disparate financial accounts for better monitoring, and those that are more focused on a particular end-goal such as increasing saving rates or assisting in paying down debt.

Works such as those by Carlin et al. (2017), Baker (2018), Baugh et al. (2018), and Medina (2021) utilize data obtained from aggregator apps. A significant benefit of these apps is that users typically have a comprehensive set of financial accounts linked, allowing for both more flexibility in the sorts of questions that can be asked and more confidence in the validity of a given set of results. In contrast, Kuchler & Pagel (2018) and Baker et al. (2022a) use data from apps that focus more on encouraging savings or debt repayment. As a result, both of these have potential issues with sample selection and also with the completeness of the individual's finances that are visible within the product.

At least one government entity has also worked to develop a transaction database through a government survey (see Schuh 2018). The Diary of Consumer Payment Choice, run by the Federal Reserve Bank of Atlanta, measures spending behavior through a transaction-level diary among a group of survey respondents. This approach combines the utility of surveys with the granularity of transaction data and makes the data widely available to researchers. However, the survey has limitations compared to the proprietary data obtained from financial institutions, given its much smaller sample size and time window (i.e., only a few days of transaction data per respondent).

2.3. Ancillary Data Linkages

The basic building blocks of transaction databases share many similarities across the large variety of sources of such data: Generally they provide a complete view of time-stamped financial flows at a transaction level for a set of financial accounts, across a large sample of individuals. This provides high-quality information regarding household consumption flows. Depending on the source, transaction data can be accompanied and enriched by other types of financial information and by demographic characteristics.

Some of the additional data take the form of account- or transaction-level metadata. Making use not only of the date of a transaction but also of the precise time of the day that a transaction takes

²For example, Xing et al. (2021) use data from Alipay to determine the impact of short-term spending subsidies on consumer spending.

place, Gu et al. (2018) identify shirking at work by examining online shopping purchases during working hours. Aggregators and other financial platforms also generally track how individuals interact with their various online accounts. Olafsson & Pagel (2017) utilize internal data from a FinTech app as a proxy for attention paid to investment accounts by measuring the number of times a user logs into their account.

Beyond the metadata from the transactions and accounts, many sources link self-reported demographic information to the individual in question, such as age, gender, marital status, level of education, and number of children (e.g., Ganong et al. 2020). This can also be combined with either self-reported or geo-tagged information on the location of residence or of each individual transaction. Whereas not all sources contain self-reported or administrative data on demographics, some of the firms take the approach of constructing proxies or best guesses for demographic characteristics (e.g., gender and age ranges) using observable financial or account characteristics.

A number of researchers are also able to put detailed location data to good use in conjunction with various observed transactions. For instance, Gelman et al. (2017) utilize locations to identify the specific gasoline prices that individuals face, whereas Meeuwis et al. (2021) use zip code-level locations to proxy for likely political leanings among users of an investment service. Agarwal et al. (2021b) focus explicitly on location-driven peer effects in Singapore as they examine how bankruptcies affect spending and other financial behaviors among coresidents of individual apartment blocks.

Another key source of data linked to transaction databases have been surveys administered by the source firm either for its own purposes or on behalf of researchers. Studies such as those by Schuh (2018), Bachas et al. (2020), and D'Acunto et al. (2020) all employ surveys of individuals to better contextualize the transactions that are directly observed.³ Such efforts can help researchers better understand the beliefs, expectations, and reasoning behind the financial decisions that are directly observable, at least partially mitigating one of the issues with context-free transaction data.

For some sources, researchers may also be able to link to various credit reporting agency information like credit scores, numbers of accounts, or negative marks on credit. Elizondo & Seira (2017) and Ponce et al. (2017) are able to link bank account data to credit bureau data in collaboration with the government of Mexico.

Transaction databases stemming from financial aggregators are some of the most complete and flexible. This is driven, in part, by the ancillary data that are linked to the financial accounts. Many aggregators contain self-reported data not only on demographics but also on financial goals and budgets, financial risk tolerances, and financial sophistication. Such data can help researchers understand how shocks can have heterogeneous responses across agents who vary greatly in key modeling parameters like risk preferences.

Access to the underlying data from FinTech apps has also enabled research into the impacts of this emerging technology itself on household financial behavior. Carlin et al. (2017) take a broad overview of the usage of a FinTech app in Iceland, examining differences in patterns of usage across generations. Becker (2017) uses transaction data from a large European bank to demonstrate that access to a money management tool encourages households to start saving and increase savings balances. Levi & Benartzi (2020) utilize the rollout of a FinTech app across different device types (e.g., Android, iOS) to establish that users pay more attention to their finances and cut discretionary expenditures, with the effects concentrated among lower-income and high spending-to-income users.

³Other researchers, such as Giglio et al. (2021), have taken this approach to map brokerage transactions to survey responses regarding the beliefs of investors.

Beyond using financial accounts or other data that are explicitly linked to the individual, many researchers have been able to make numerous inferences about individual preference characteristics using transaction data. For example, a number of researchers identify a measure of financial sophistication through the propensity for individuals to make financial mistakes like overdrafting or paying credit card penalties when possessing sufficient liquidity to avoid them (e.g., Carvalho et al. 2019, Jorring 2019). Attention to and salience of financial data and accounts are measured with the frequency by which individuals login to their accounts, and labor market status can be deduced from the receipt of regular direct deposit paychecks or social security income. Account balances may even be partially imputed (when lacking account balance data) using interest income in bank accounts. These types of inferences are continually refined and augmented across a number of transaction data sources as more researchers begin to work with this class of data.

3. COMPARISON WITH ALTERNATE MEASURES OF HOUSEHOLD FINANCES

Measuring consumption using household transaction data offers many benefits but also some downsides compared to the use of other sources of household financial data. **Tables 1** and **2** offer a comparison between the attributes of transaction data and those of more traditional sources of data generally used for measuring household consumption. **Table 1** offers a broad comparison between the attributes of transaction data and those of more traditional data sources (e.g., government surveys or administrative data). **Table 2** notes variations in attributes that are driven by the source of transaction data (e.g., payment processor, bank, or FinTech firm). Importantly, the relative weights placed on these attributes can vary depending on the nature of the question the researcher is asking; each class of data has situations in which it dominates.

Research requiring product-level consumption data will necessitate something akin to scanner data or detailed consumer diary surveys, which describe consumption at a more granular level compared to the store-level transactions available from transaction data. Questions revolving

Table 1 Attributes of consumption data sources

Properties/data source	Transaction	Survey	Administration	Scanner
Examples	JPMCI, Mint.com	PSID, CEX	Norwegian Tax Registry	Nielsen Consumer Panel
Frequency	Very high	Often quarterly	Often annual	High
Measurement error (\$)	Low	Medium	Low	Low
Measurement error (category)	Medium	Low	High	Low
Sample size	High	Low	Very high	Medium
Comprehensiveness	Medium to very high	Medium to high	High	Low
Demographics	Medium	High	Low	Medium
Geographic location	High	Low to high (confidential)	High	Medium
Representativeness	Varied	High (falling; tail issues)	Very high	Medium
Historical coverage	Low	High	Medium	Low
Panel length	Medium (growing)	Typically low	High	Medium
Granularity	High	Medium	Low	Very high
Asset coverage	Low to very high	Medium	High	Low

Abbreviations: CEX, Consumer Expenditure Survey; JPMCI, JPMorgan Chase Institute; PSID, Panel Study of Income Dynamics.

Table 2 Attributes of sources of household transaction data

Properties/data source	Banks	Credit cards	Neutral aggregators	Focused aggregators	Payment apps
Example	JPMCI	Visa	Mint.com	ReadyForZero	Alipay
Sample size	High	High	High	Low	Medium
Account linkages	High	Low	Very high	Medium	Low
Observable income	Yes	No	Yes	Yes	No
Representativeness	High	High	Medium	Low	Low
External links and surveys	Medium	Low	Varied	Varied	Low
Legal access hurdles	High	Very high	Medium	Low	Medium

Abbreviation: JPMCI, JPMorgan Chase Institute.

around durable purchases and financing may be difficult to understand using transaction data, as such purchases often appear in bank accounts as a stream of financing charges rather than as an actual purchase. Here, data from tax registries or credit bureaus may dominate. Transaction data will also be inferior when looking at aggregate consumption trends over time, since data predating 2010 are generally not available from such sources.

3.1. Advantages of Transaction Data

Some of the longest-standing measures of consumption available to researchers are government-run surveys such as the Consumer Expenditure Survey in the United States and the Living Costs and Food Survey in the United Kingdom. However, surveys are limited in scope and complexity due to both monetary and time costs. Individuals cannot be asked to answer every survey question about types of spending or sources of income that might ever be of interest to researchers. Data collection in transaction databases is, by contrast, fully automated. Adding an additional transaction incurs no time cost for users and requires minimal monetary costs to be logged and stored in the database in perpetuity.

Unconstrained by marginal cost concerns on the part of both the user and the entity collecting the data, every individual transaction can be stored along with metadata regarding that transaction. Therefore, while a survey question cannot easily be asked retroactively, the granular and complete nature of transaction data at their time of generation means that they can be mined far in the future for indicators of behaviors or purchases. This is often made possible by the extreme detail offered by the transaction description field, as is elaborated on in Section 4.5.

Assuming a complete view of all of an individual's accounts, the values of spending, income, and assets are exact and are not as prone to recall bias as is the case for a survey result. This exactness can be especially useful when working to measure the value of highly volatile assets like equity portfolios or when investigating atypical spending patterns.

Completeness across financial accounts also drives the main benefits of the so-called scanner data such as those of the Nielsen Consumer Panel (NCP), which tracks subsets of household purchases at an individual product level. Scanner data can grant even finer detail on elements of household spending by decomposing a transaction into its constituent products. However, the set of covered products and stores is limited (generally to nondurable purchases from retail stores), and other financial attributes of the individual are often lacking.⁴

⁴The NCP covers approximately \$400 of household spending per month for its panel members, a small fraction of total household spending in the United States.

Table 3 Sample sizes

Paper(s)	Data source	Sample size and composition
Various	Current Population Survey	60,000 households
Various	Consumer Expenditure Survey	7,000 households
Various	Nielsen Consumer Panel	40–60,000 households
Various	Living Costs and Food Survey	6,000 UK households
Carvalho et al. 2021	BBVA	6,300,000 individuals
Baugh et al. 2018	Large US aggregator app	2,700,000 individuals
Andersen et al. 2020d	Danske Bank	1,100,000 individuals
Bachas et al. 2020	Bansefi	348,802 individuals
Agarwal et al. 2021b	Leading bank in Singapore	180,000 individuals
Ganong & Noel 2019	JP Morgan Chase Institute	182,361 households
Kim et al. 2020	JP Morgan Chase Institute	380,000 businesses
Baker & Yannelis 2017	Large US aggregator app	152,810 government employees
Olafsson & Pagel 2018	Meniga aggregator app	65,000 Icelandic households

Sample sizes obtained from summary statistics tables within the listed papers. Survey data sample sizes are approximate. Abbreviation: BBVA, Banco Bilbao Vizcaya Argentaria.

Similarly, researchers employing data from credit bureaus can track some aggregated versions of household spending through the observation of credit card balances. Durables purchased with debt, such as vehicles and homes, are well tracked in such data, but credit card balances mix revolving debt with current charges (see Fulford & Schuh 2020), making it difficult to ascertain the flow of household consumption even when all household purchases are made via credit card rather than debit or cash spending.

Although the data available from other databases may have specific benefits, transaction data often have an advantage over other data sources due to their sheer size. **Table 3** enumerates the sample sizes of some common survey databases as well as transaction databases used by researchers in recent years. The sample sizes of transaction databases are often one or two orders of magnitude larger compared to other data sources, yielding correspondingly smaller standard errors, especially when examining the behavior of a particular subset of the population.

This size advantage has fueled a substantial expansion of research into the micro foundations of the macro economy and has focused new attention on models that incorporate heterogeneity across individuals and households. The ability to divide a sample jointly by income, liquidity, credit, and age can help test new tools such as Heterogeneous Agent New Keynesian (HANK) models, which add both heterogeneity and idiosyncratic risk to Representative Agent New Keynesian (RANK) models. More broadly, new questions about inequality of wealth and income can be more thoroughly answered when the top 1% of a sample is composed of 10,000 individuals rather than 500.

Recent years have also seen an increase in the use of detailed administrative data on income and wealth to impute consumption as the residual of the household's budget constraint. Governments of countries such as Denmark, Sweden, and Norway have made annualized financial data on all citizens available for research, including bank balances, income, and asset holdings. The use of these sources is exemplified by the studies of Koijen et al. (2014) in Sweden, De Giorgi et al. (2020) in Denmark, and Fagereng & Halvorsen (2017) in Norway. Relative to transaction-level data, these databases substantially reduce issues of sample selection and incompleteness while also making other links to government databases possible. However, they do not provide any high-frequency measures of financial activity and generally cannot measure spending directly, as only an aggregate

annual level of spending or consumption can be imputed. In addition, calculating consumption as a residual implies that different types of consumption (e.g., spending on nondurables, food, services, or durables) cannot be observed.

In developing countries with fewer resources to devote to systematically collecting economic data, transaction data can also fill a gap in coverage. Although not all countries collect detailed survey-based micro data, all have substantial penetration of banks, card payment systems, or FinTech products. Just as many countries are leap-frogging wired telephone infrastructure in favor of cell phones, some countries have largely bypassed credit and debit cards, moving to mobile-based payment systems. In such areas, collaborating with FinTech apps can give researchers and policy makers a clearer view of the economy through this type of transaction data.

Finally, transaction data increasingly allow researchers to understand not only what has happened but also what is happening in the present. As seen most clearly in the proliferation of research employing transaction data during the COVID-19 pandemic, transaction data can be utilized to give policy makers a new and highly detailed view into the current state of the real economy, beyond the financial markets, without having to wait weeks, months, or years for surveys to be administered.

The high-frequency nature of the data allows researchers to unpack fast-moving events like a government shutdown (e.g., Gelman et al. 2020) or the onset of the COVID-19 pandemic (e.g., Chen et al. 2021), or even to examine patterns of spending across hours of the day (e.g., Gu et al. 2018). This enables researchers to observe economic trends and shocks directly rather than imputing effects from data aggregated at the monthly, quarterly, or annual level.

3.2. Limitations of Transaction Data

Household transaction data offer a large number of advantages over other sources that also measure household financial activity. However, this class of data features drawbacks, too, which mostly stem from the hands-off nature of data collection.

3.2.1. Account completeness. Although transaction data offer a number of substantial advantages over more traditional data sources, they have disadvantages as well. One basic difficulty is interpreting the unit of analysis because, for many such databases, the unique identifier is an account. Whereas in a survey a surveyor can set well-defined boundaries on the unit of analysis, households often use individual financial accounts and aggregator apps in a variety of ways. However, each member of a married couple may have their own account with the financial aggregator app that links to both their individual and joint accounts; or they may have a single account with the app that contains both individual accounts as well as all the joint accounts.

In general, a researcher can never be certain that they are observing all the relevant financial accounts owned by an individual or a household, and it is necessary to make assumptions or sample restrictions to try to isolate the more complete and active users. Many researchers take the approach of excluding users with small numbers of transactions per month, no visible sources of income, or financial transfers to accounts that are unlinked. Such ambiguity may make it difficult to interpret the raw levels of income and spending and compare them with other sources of information where the unit of analysis is more clearly documented. More broadly, these databases tend to lack the decades of supporting documentation found in government surveys and other sources that are designed from the ground up to be used by researchers and to make accurate economic inferences.

3.2.2. Sample selection. Even in the most information-rich and comprehensive transaction databases, some disadvantages remain. One common criticism is that, by their very nature,

financial transaction databases are composed of a selected, rather than a nationally representative, sample of individuals. Even when dealing with data sourced from large national financial institutions, there exists a sizable minority of the population that do not have bank accounts or credit cards of any type and deal primarily in cash.

The selection issue becomes more acute when the transaction data are obtained from more specialized financial institutions or FinTech apps that are targeted at particular slices of the population. Some apps may be aimed at poorer individuals who have trouble saving, at individuals who enjoy trading stocks, or at individuals who have a variety of financial accounts that need to be managed. Each of these data sources may exhibit some bias relative to the targeted population when investigating a given question. Baker (2018) uses the demographic and financial characteristics linked with transaction data to demonstrate that such sample selection issues can be largely resolved using population weights and given sufficient numbers of users.

3.2.3. Transaction categorization. Spending done via cash or paper checks can pose problems for researchers using transaction data. Cash transactions cannot be observed directly, though they may be inferred from ATM withdrawals. Check transactions can be observed but cannot be automatically categorized into the types of spending that they represent. This concern has declined as cash and checks have come to represent smaller fractions of spending, though the growth in peer-to-peer cash transfer apps (e.g., Venmo) often raises some of the same concerns.

More broadly, unlike scanner or some types of survey data, categorization of spending is necessarily done at a transaction level rather than at a product level. While a transaction at a restaurant is easily interpretable, a transaction at a big box retailer (or an online retailer like Amazon) may contain a range of products, from nondurable consumables like groceries to semidurables and durables like clothing, furniture, or mechanical equipment. Researchers may choose to focus on more unambiguous categories such as restaurants when, for instance, they discuss the responses of nondurable spending to an income shock.

Categorization of spending within household transaction data is generally done by algorithms constructed by the parent organization collecting the transaction data (e.g., the bank or account aggregator). Typically, these utilize a combination of Merchant Category Codes (MCC) linked to merchant IDs and textual analysis of transaction descriptions to arrive at a category for a given merchant. In some cases, miscategorizations may occur or transactions may be left uncategorized. However, the textual descriptions and other metadata linked to each transaction can allow researchers to improve on such categorization algorithms or to directly observe particular transactions of interest without being limited to the categories assigned by the parent organization.

3.2.4. Replication. The proprietary nature of these databases can present problems in terms of accessibility for future research and of the ability to replicate a given analysis. In an extreme case, some private sources of data such as smaller FinTech apps may go out of business or deprecate older data to minimize storage costs, making it impossible to replicate results. More broadly, research done with some of these databases is the result of personal relationships between researchers and the firms, and there is often no established way for other researchers to gain access.

These problems do not affect many of the traditional data sources that are primarily sourced from public agencies. Many common government survey data are publicly accessible online or have a standard application process for gaining access to more detailed data. One avenue for expanding the utility of transaction databases is for governments or large research institutions to cement more permanent means of collaboration between researchers and private data providers.

4. RESEARCH AVENUES EXPLOITING BENEFITS OF LINKED ACCOUNTS

4.1. Micro-Based Aggregate Consumption Statistics

When it comes to tracking consumer spending in the economy, many economists have turned either to national aggregates such as the Personal Consumption Expenditures (PCE) from the National Income and Product Accounts (NIPA) or to national surveys such as the US Census Bureau's Retail Trade Survey. However, these data are produced with a substantial lag and provide no detail at a subnational level. Using transaction data as a building block for better aggregate statistics offers a promising path to a clearer picture of the economy.

Batch et al. (2019) test whether modern machine learning paired with alternative (and faster-to-update) data can provide better advance GDP estimates or lower the size of GDP revisions. Although they only have access to an aggregated version of transaction data provided by a financial services vendor, they find that such data are extremely valuable for better predicting components of national accounts. Aladangady et al. (2019) go further, using transaction-level data from a large electronic payments company to provide a disaggregated and higher-frequency version of these national consumption statistics. They demonstrate the utility of these data in tracking spending during fast-moving economic shocks like major hurricanes in 2017 and the 2019 government shutdown.

Of these various efforts, those by Chetty et al. (2020) are perhaps the most ambitious. In the authors' own words, they "build a publicly available platform that tracks economic activity at a granular level in real time using anonymized data from private companies." Many of the statistics presented and tracked are derived from detailed transaction data that allow the authors to provide many dimensions of heterogeneity in high-frequency spending behavior, including disaggregated statistics by geographic area, income, and industry. The ability to provide such detailed statistics publicly in a continuous and up-to-date format is of great importance to both researchers and policy makers to understand economic events as they occur rather than after months of waiting for reliable data.

4.2. Linked Financial Accounts

Household transaction data by themselves allow for a wide range of new insights into household behaviors and responses to economic shocks. Einav & Klenow (2021) and Baugh & Correia (2022) work solely with transaction databases that do not feature much other information about account balances, assets, or demographics.

However, while transaction data have proven useful even when unaccompanied by a fuller suite of financial information, they have had the largest impact when paired with more complete household financial information. Given that the sources of the data are most commonly banks and financial aggregators, it is common to have comprehensive overviews of an individual's financial information across a range of account types. One of the most important components that such financial data offer is a time-varying measure of individual liquidity and credit access. This has allowed a large group of researchers to demonstrate the importance of liquidity in driving household responses to both anticipated and unanticipated income shocks. Agarwal & Qian (2014), Kueng (2018), Ganong & Noel (2019), and Andersen et al. (2020c) all leverage metrics like account balances and credit availability to map out household-level heterogeneity in consumption behavior.

Agarwal & Qian (2014) and Baker (2018) apply these data to the classic questions about the consumption response to unanticipated income shocks and how it varies across households with different levels of liquid assets and credit access. Agarwal & Qian (2014) use transaction-level

data from a Singaporean bank to investigate the consumption response to an economic stimulus policy in Singapore; they find that households begin to respond upon the announcement of the upcoming payments and that households with lower levels of liquidity and credit respond more strongly to it. Baker (2018) uses shocks to households' employers to instrument for permanent income shocks, finding similar levels of heterogeneity across credit access and liquidity.

Another common linkage is between transaction accounts and home mortgage accounts, which enables researchers to examine how home values, mortgage terms, and mortgage performance interact with consumption decisions. Park (2016) takes advantage of information from household transactions to predict mortgage delinquencies, using the propensity of households to being charged late fees or overdrafts as a proxy for risk. Similarly, Jorring (2019) shows that mistakes in checking or credit accounts (e.g., avoidable overdrafts or late fees) can predict the consumption response to predictable changes in mortgage payments. Households that make such financial mistakes often fail to smooth consumption around these predictable events.

A number of papers, such as those by Bernstein (2021) and Ganong & Noel (2020), have been able to examine directly the impact of home values, mortgage policies, and mortgage terms like Adjustable-Rate Mortgage (ARM) resets on household consumption behavior and labor supply. Given the prominence that the costs of homes and mortgages have on household balance sheets around the world, such questions are of first-order importance when thinking about the transmission of monetary policy or associated changes to federal home loan policies.

Beyond houses and mortgages, linking transaction data to investment and retirement accounts has also provided more clarity about household balance sheets. Brauer et al. (2019) and Meyer & Pagel (2019) look to better characterize consumption responses to dividends and to both realized and unrealized capital gains and losses. Again, transaction data provide a way to identify credibly how asset price fluctuations may be transmitted through the economy, and they can also highlight heterogeneity across owners of these assets.

Other researchers work with transactions across multiple financial accounts to test for offsetting behavior that may net out in the aggregate or may be missed if observing only one type of financial accounts. Medina (2021) highlights the importance of testing for offsetting financial behavior across separate financial accounts. Garcia et al. (2021) examine whether similar offsetting behavior comes about after inducing individuals to save more through a similar intervention, but they find little change in credit utilization. Kuk & Raaij (2021) look at savings distributions across individual and joint accounts within households, finding a wide range of joint allocations.

4.3. Evaluating Heterogeneity and Models of Household Behavior

Models created to understand and rationalize household behavior, especially consumption behavior, can be more thoroughly tested when increased sample sizes are combined with the extreme detail allowed for by transaction data. Key parameters of these models, like risk aversion and salience, can be more directly captured, and important financial characteristics such as the precise levels of liquidity and credit access are readily observable. Ponce et al. (2017) note that relative interest rates are poor predictors of how households make decisions about which cards to utilize and which ones to pay down. However, in an experiment that randomized the temporary interest rate offers when opening a new card, they find that households responded to these more salient price changes, suggesting models of limited attention or mental accounting.

The sensitivity of consumption to beliefs, news, and income shocks of various types has generated a large array of research using transaction data. Most of these papers address models of intertemporal optimization, often testing various aspects of the permanent income hypothesis or of buffer-stock models. Transaction data at the individual or household level are invaluable to such

efforts, as many of the common explanations for consumption fluctuations revolve around income volatility or uncertainty, levels of liquidity and wealth, spending shocks, and durable spending. Linking detailed transaction data with financial account balances and credit information, these explanations can be tested jointly and with great precision across large sample sizes. Moreover, identifying precise income shocks, as well as understanding just how anticipated or unanticipated those shocks were, is made much easier when each individual income transaction can be observed and fully characterized according to its source.

As one example, Ganong & Noel (2020) link transaction data with mortgage data to assign mortgage defaults to one of two categories: strategic defaults, if households default when the debts are too high relative to their home values, and liquidity-driven defaults, if the monthly payments are too high given the current household resources. They find that the overwhelming majority of defaults is liquidity- or adverse event-driven rather than strategic, and this evidence can inform debates over optimal mortgage and debt relief policies.

Looking at responses to macroeconomic news events, Garmaise et al. (2020) use household transaction data to identify persistent excess sensitivity of consumption, even when such events are not necessarily meaningful for the household. For more salient events—i.e., those that garner more media attention—responses are higher, as are responses among lower-income households.

Other papers address the response of households to personal income shocks. Olafsson & Pagel (2018) highlight excess spending among individuals on their paydays among most of the population of Iceland. They show that even for high-income and highly liquid individuals, paydays tend to induce appreciable spending responses across many categories of consumption, despite the fact that paydays are known in advance. They explicitly test multiple models of household behavior and reject the possibility that either current or imagined future liquidity constraints are the drivers of such spending responses. Olafsson & Pagel (2019) turn to unanticipated income shocks from lotteries, finding marginal propensities to consume (MPCs) above 1, financed by borrowing, for small wins and low MPCs for large wins, again pointing away from liquidity constraints as the main driver of large MPCs out of windfalls.

Households have shown excess sensitivity to high-profile and perfectly anticipated payments from the Alaska Permanent Fund, as well. Kueng (2018) notes that, in this case, excess sensitivity increases with income and is high even for those with high levels of liquid assets. Although this behavior stands in contrast to most classical rational models, he shows that the actual welfare loss from such non-smoothing behavior is exceedingly small for these households, suggesting that many consumption deviations identified by other researchers are also likely minimal from a welfare perspective.

Gelman (2021a) provides further evidence showing that liquidity constraints are not the primary driver of excess sensitivity of consumption. The author tests a range of parameters of a canonical buffer-stock model and notes that a quasi-hyperbolic version of the model performs better than the standard exponential formulation in explaining the observed levels of excess sensitivity. Gelman (2021b) takes advantage of the vast sample size afforded by transaction data to decompose the aggregate or average MPCs across individuals with varying observable and unobservable characteristics. He returns to a buffer-stock model and provides perhaps the first clear data-driven breakdown of the importance of these persistent characteristics and time-varying shocks.

Heterogeneity across individuals is also the prime focus of Kuchler & Pagel's (2018) work. The authors demonstrate that individuals often fail to stick to explicit plans to pay down their debt, which is consistent with the present-biased agents who feature in models with quasi-hyperbolic discounting. Using detailed transaction data, they further show that heterogeneity in debt repayments across individuals is strongly correlated with cross-sectional variation in short-run impatience and financial sophistication.

Gathergood et al. (2019) and Gathergood & Olafsson (2020) use transaction data to test theories of mental accounting at the household level. Gathergood et al. (2019) find that households do not treat all credit card expenditures equally; rather, they tend to pay down debt incurred by certain types of expenses more quickly than others. In particular, households tend to pay down debt more quickly for nondurable expenditures relative to durable expenditures. Agarwal et al. (2021a) examine how household spending changes in the days around credit card statement dates. They suggest that household spending tends to rise substantially following these statement dates, but that most of the variation can be rationally accounted for by the fact that households take advantage of the free float that credit cards offer until the following pay period.

Lastly, Gathergood & Olafsson (2020) turn to the puzzle of households that co-hold liquid balances alongside high-interest debt. Using linked transactions and credit and bank balances, they calculate these propensities at a daily level, finding that instances of co-holding are usually very short term and that rational explanations are likely subordinate to mental accounting.

4.4. Experimental Interventions

Transaction data have also been utilized in conjunction with experiments targeting households. With such detailed data, gathering posttreatment data is automatic and comprehensive, allowing for larger samples and more detailed investigations.

As one example, Aydin (2022) randomizes the extension of credit lines to customers of a large Turkish bank, finding substantial propensities to borrow even among those with sizable assets. He takes advantage of the completeness of the credit, income, spending, and asset data to build support for a buffer-stock model rather than more behavioral explanations. D'Acunto et al. (2020) also investigate the spending responses to the provision of a new credit facility for borrowers, eliciting preferences through the FinTech app from which they obtain transaction data. With these data, they are able to fully characterize spending responses across both the income and liquidity distributions as well as across users with various beliefs and risk preferences.

Such linked data can also offer convincing evidence of a role for behavioral mistakes in household behavior. Medina (2021) runs an experiment on users of a financial account aggregator in Brazil, randomly sending reminders to pay credit card bills to users of the service. She finds that households significantly increase on-time payments but also increase the probability of overdrafting on other accounts they own. This finding highlights one strength of transaction data stemming from financial aggregators: Researchers can easily observe potentially offsetting behavior in other financial accounts owned by the same individual in response to a shock.

4.5. Utilization of Transaction Text Fields

One of the most flexible components of the common classes of transaction data thus far used by researchers is the text field, that is, the textual description that accompanies the amount and time stamp for each transaction. This text (or a simplified version of it) is what is observed when one examines their itemized credit card statement at the end of a billing cycle. Although this text may feature complex numerical identifiers or name abbreviations, it is generally possible to use it to determine which firm a particular expenditure transaction was paid to or which employer is the origin of a particular income transaction.

This field thereby allows a researcher to pick out transactions that may signal an employment relationship, the timing of a government benefit receipt, spending at a particular retailer, or the occurrence of an event of interest. These relationships can all be discerned without explicitly asking a household about any of these linkages or events, thus giving researchers the ability to extend their analysis years back in time.

Baker (2018) uses direct deposit transaction descriptions to identify individuals' employers and then measures the unanticipated changes to households' income that stem from shocks to the employers. Given the scarcity of data linking employees and employers, the ability to generate these links is extremely valuable for understanding the transmission of shocks through the economy. Gelman et al. (2020) and Baker & Yannelis (2017) also utilize transaction text fields to identify households that contain employees of the federal government in order to analyze differential spending behavior when such households were subjected to missed paychecks as part of the 2013 federal government shutdown.

Others, such as Agarwal et al. (2021c), Einav & Klenow (2021) and Baker et al. (2020a), have used a similar approach to map transactions to the numbers of firms that individuals shop at, mostly retailers or consumer service firms. Linking to a broad group of firms enables these authors to analyze the firms' customer bases and to gain more information on risk to firm growth, and it adds the ability for the researchers to disaggregate sources of firm revenue. Baugh et al. (2018) focus on transactions at a single firm, Amazon, and examine responses to changes in sales tax policy. Such linkages to spending made at individual firms is not possible using alternative consumption data sources like government surveys and scanner data or by imputing consumption via tax registries.

Another common tactic when using transaction data has been to identify less typical sources of income or benefits. Although administrative data may generally be required to observe the precise amounts of unemployment insurance paid out to individuals, Ganong & Noel (2019) and Andersen et al. (2020c) are able to pick out those payments using the textual descriptions within transaction databases. Karger & Rajan (2020) take the same approach with CARES Act stimulus payments in 2020, Kueng (2018) with payments from the Alaska Permanent Fund, and Olafsson & Pagel (2019) with winnings from national lotteries. Baugh (2015) and Carvalho et al. (2019) pick out loans from and repayments to payday lenders, and Gelman et al. (2019) identify the precise timing and amount of tax returns deposited into accounts via direct deposit.

Overall, the text field found in many types of transaction data offers researchers unparalleled room for creativity in identifying particular sources of income or types of spending.

4.6. Transaction Data During the COVID-19 Pandemic

A unique demonstration of the utility of transaction data came during the COVID-19 pandemic that first swept through the world in 2020. This pandemic brought about a deep recession in many countries that was notable not only for its size but also its speed. The rapidity of the economic shocks was unlike anything seen in modern history, with many countries experiencing the largest-ever single-month or single-quarter declines in employment, spending, and GDP.

The speed at which this event occurred means that governments were grappling with policy proposals for household and business support without being able to observe many of the economic indicators that are only produced with a lag of weeks or months. Turning to transaction data, many researchers were able to quickly produce diagnoses of the size and distribution of consumption and income declines. Such analyses were conducted in a wide range of countries—for example, by Andersen et al. (2020a) for Denmark and Sweden, Chetty et al. (2020) and Cox et al. (2020) for the United States, Chen et al. (2021) for China, Carvalho et al. (2021) for Spain, Chronopoulos et al. (2020) and Hacıoglu et al. (2020) for the United Kingdom, and Bounie et al. (2020) for France.

Governments and central banks often focus on the behavior of financial markets to provide high-frequency views on economic activity, especially in response to recent policy initiatives. The newfound flexibility and utility of transaction-based consumption data during a fast-moving crisis may prompt governments and other institutions to set up persistent indicators of consumption leveraging high-frequency transaction data. If these consumption indicators are made available in the future, they may help policy makers to calibrate countercyclical policies.

5. FUTURE DIRECTIONS OF RESEARCH

5.1. Analysis of Firms Using Transaction Data

An emerging research agenda has begun to leverage households' financial transactions to study features of firms and the risks that they face. Researchers are able to map individual households' transactions to particular firms using either unique firm identifiers or the text fields from credit and debit transactions. Novel firm-level panels detailing revenue sources and customer base attributes can thus be constructed from household transactions.

Agarwal et al. (2021c) use credit card data to produce disaggregated firm-level sales data and find that such data are predictive of future stock returns. Moreover, they find that the composition of customer bases is important, with stronger stock returns following increases in spending from customers with higher credit scores, wealth, and loyalty.

Baker et al. (2020b) demonstrate that aggregated firm-level statistics derived from household transaction data match well to traditional sources of firm-level data like Compustat. They develop two novel statistics about customer bases that cannot be generated using traditional data sources: the rate of churn in a firm's customer base and the pairwise similarity between two firms' customer bases. They then demonstrate that these statistics can be used to inform alpha-generating trading strategies as well as to provide new insights into firm investment and markups.

Finally, Einav & Klenow (2021) use debit and credit card transactions spanning the Visa network to demonstrate the importance of measuring customer bases when working to understand firm growth, and they demonstrate that extensive margin adjustments (e.g., customers leaving or joining a customer base) are responsible for the vast majority of firm-level sale variations over time.

Although this window into firms provides substantial additional value to these household transaction data sets, there are limitations to this approach. Because the source of the data is ultimately a database of consumer transactions, firms that obtain sizable amounts of revenue from other businesses or governments will be poorly observed or measured. Moreover, only firm revenue, not costs or employment and production characteristics, can be directly observed. Despite these drawbacks, consumer-facing firms constitute a large portion of most economies and feature prominently in questions spanning fields like finance, marketing, and industrial organization.

5.2. Analysis of Firms Using Firm-Level Transaction Data

Another promising area of research leverages expanded sources of transaction data to directly observe firm-level transactions. Much as transaction data have enabled a more thorough analysis of household behavior and heterogeneity, the transactions made by tens of millions of small businesses can increase visibility into this large sector of the national economy. Especially in countries with less reliable measures of small business activity, the ability to observe transactions from these firms directly, by working with banks or other financial intermediaries, can greatly empower researchers and policy makers.

Pioneering work in this field was made by Weber (2019) and Kim et al. (2020). Weber (2019) links the financial transactions of millions of individuals at a large financial institution to the universe of small business transactions at that same institution. The author focuses on the founders of small businesses, measuring personal financial characteristics such as cash flow management abilities, which are observable from the founders' personal accounts. He notes that such financial management skills are highly predictive of the financial success of the small businesses subsequently founded by these individuals.

Kim et al. (2020) work with the JPMorgan Chase Institute to gain access to transaction-level data from hundreds of thousands of small businesses. In particular, they are able to match the

transactions of small businesses to the personal accounts of those businesses' proprietors and to map out how the fortunes of the businesses impacted proprietors' consumption during the COVID-19 pandemic.

5.3. Integration with Governments

Transaction data have the potential to make substantial contributions to future research and policy through additional integration with governmental agencies and central banks. Current research using transaction data often involves one-off relationships between researchers and data providers. For a range of reasons that encompass both competitive and legal concerns, only a small number of firms have allowed for more regular access to their transaction databases. As a result, obtaining and maintaining access to such data is difficult or impossible for most researchers. Further, the small scale of most collaborations requires learning the idiosyncrasies of each particular transaction database.

Partnerships with government agencies may provide one way to create a more standardized process for access; they may also leverage the expertise that many agencies have built up in producing credible and well-documented statistics regarding national product accounts or other economic indicators. Oversight of data access by federal regulators, built over years of facilitating access to tax or census records, may also provide the legal assurance firms need to ensure that vetted researchers will not misuse their data.

In addition, linkages of transaction data to other sensitive economic databases, such as tax or credit registries, could be facilitated. In Denmark, Andersen et al. (2020d) use transaction data matched to government registry data to measure the extent to which informal transfers from family and friends act as an insurance against negative income or labor supply shocks. In the United States, Chetty et al. (2020) work with a large number of private sector partners to build a database of economic indicators that is both granular and timely. Overarching sponsorship or organization of such efforts by a neutral party like the Federal Reserve could provide an avenue to expand these efforts.

Governments would be well served by aiding in these efforts in systematic ways. As demonstrated during the COVID-19 pandemic, transaction data can provide some of the most up-to-date statistics and can serve to inform policy decisions at a much finer level than the aggregate statistics used in the past. In addition, as governments around the world tackle persistent questions of inequality, transaction-level data offer an open window into the economic conditions and decisions of millions of households across the income distribution.

6. CONCLUSION

The past decade has seen a tremendous amount of growth in research utilizing a new class of data centered on individual financial transactions. These data are often linked to other financial data such as asset balances, credit, and equity holdings as well as self-reported demographic characteristics and geographical information. Whereas in the past many of these detailed aspects of household finance were available only for a small slice of the population or had to be imputed from less granular survey questions, the financial transaction data available today make most financial and labor market attributes of a household directly observable.

These databases have enabled researchers to use new identification strategies to answer more decisively classic economic questions and to obtain more precise empirical estimates. Such questions include how households respond to economic shocks, how government monetary and fiscal policies are transmitted to households, and how important household balance sheets are to broader economic trends.

Beyond allowing researchers to revisit these questions, this class of data has also proven to be one of the most flexible in terms of potential for broader-scope research. Researchers have taken advantage of the massive sample sizes to highlight the importance of heterogeneity in both economic means and economic beliefs and behavior in explaining how the economy moves in response to aggregate shocks. The ability to use the textual transaction descriptions to identify ex-post particular firms, income sources, and detailed types of spending has enabled researchers to think creatively about particular shocks to individuals or to define multidimensional characteristics that can be important for structural estimation.

The speed of analysis made possible by these data has also been critical during the COVID-19 pandemic, as it has allowed both economic fundamentals and the response to government fiscal interventions to be tracked nearly in real time and with unprecedented granularity. This powerful demonstration seems likely to cement transaction data as an important tool for policy makers and may aid in the construction of more accessible data sets of this type for researchers.

As the number of sources of transaction data continues to grow around the world, many worries—about representativeness, the proprietary nature of many of the data sources, and the ability to replicate research—are declining. In addition, researchers from other fields are beginning to recognize the value and flexibility of such data, leading to their increased use in fields like marketing, industrial organization, development, asset pricing, and corporate finance.

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